

Lesion implementations

Lesion	Implementation	Status	Result or work remaining
Uniform amplitude loss	All RGC gains $\times$ 0.5	Completed; original tuning campaign pre-mtMix, letter tested with current mtMix	Original speed-tuning peaks fell 35–49% and coherence fell 9–18%, while tuning shape changed little. In the current two-stream letter test at 1°/s, d' fell only 4.510→4.423 (–0.087). Normalization absorbs most of the deterministic loss.
Random amplitude loss	Independent spatial gains from $U(0.3, 0.7)$	Completed pre-mtMix	Produced a coherence reduction similar to uniform loss, approximately 9–18%. Two-stream consolidated matrix still to be completed.
Patchy amplitude loss	Spatially correlated gain map, Gaussian $\sigma = 3$ px	Completed pre-mtMix; later tested with Site-2 noise	Deterministically similar to uniform loss. It also failed to separate from uniform loss under the tested V1 and MT Site-2 noise conditions.

Uniform delay	All RGC temporal kernels shifted by 2 frames	Completed, including current mtMix letter test	Approximately 0% effect on steady-state tuning. Letter d' at 1°/s changed 4.510→4.536 (+0.025). A latency/VEP observable is still required to measure its intended effect.
Random spatial delay	Independent per-pixel delay from {0,1,2,3} frames	Completed, including targeted mtMix follow-up	Pre-mtMix coherence fell 59% derivative/ 39% lagged. With mtMix: low-pass 0%, band-pass -35%, high-pass -77%. Letter d' at 1°/s increased 4.51→4.88. Does not explain the clinical slow-speed deficit.
Patchy spatial delay	Correlated spatial delay map, Gaussian $\sigma = 3$ px	Completed pre-mtMix	Behaved approximately like uniform delay and caused little steady-state disruption. Consolidated two-stream matrix still to be completed.

Uniform amplitude + uniform delay	Gain $\times$ 0.5 and all kernels shifted +2 frames independently	Completed 2026-08-31 with mtMix	At 1°/s: healthy 4.510, amplitude 4.423, delay 4.536, both 4.449. DELAY_NULL: YES; BOTH_IS_AMP: YES; SUPERADDITIVE: NO. Adding uniform delay did not worsen the amplitude deficit.
Coupled amplitude and delay	Spatially low amplitude tied to spatially high delay	Completed pre-mtMix	This is different from the independent uniform combination above. It still requires inclusion in the consolidated two-stream matrix.
Parasol-only amplitude loss	Parasol gain $\times$ 0.3	Completed on invalid older MT	Existing MT output is not biologically interpretable because the old MT was midget-dominated. Must be repeated using the two-stream MT.

Midget knockout	Midget gain = 0	Completed with two-stream MT	Speed-graded loss: approximately –61% for the slow/static tier, –16–17% near 5°/s, and –1–3% for the fastest tier. Letter loss was 0.210 d' at 1°/s and 0.059 at 5°/s.
ON-only delay	ON classes shifted by 1 frame	Completed on invalid older MT	Existing MT output is not biologically interpretable. Must be repeated through the two-stream MT.
HF exponential low-pass, $\tau = 2$	Temporal low-pass; matched amplitude $k = 0.817$	Completed first look	Increased MT d' by +0.353 at 1°/s and +0.082 at 5°/s. It did not produce high-frequency failure.
HF exponential low-pass, $\tau = 8$	Stronger temporal low-pass; matched amplitude $k = 0.580$	Completed 2026-08-31	Still increased d': +0.294 at 1°/s and +0.105 at 5°/s. HIT_MT: NO; STILL_HELPs: YES. No further exponential- τ tests are planned.

Passband-unity HF high-cut	Fourth-order Butterworth, $H(0)=1$, cutoff 0.05 cycles/frame; matched amplitude $k = 0.684$	Completed 2026-08-31	At $1^\circ/\text{s}$, d' increased $4.510 \rightarrow 4.947$ (+0.436). At $5^\circ/\text{s}$, it fell $5.309 \rightarrow 5.015$ (−0.294). This is a real fast-speed cost, but it affects the wrong clinical end. The HF-kernel family is now considered complete.
Trial-varying conduction jitter	Random temporal delay redrawn across trials	To be completed	Predicted to be more damaging than a fixed delay_random map. It should be evaluated using mean d' and trial variability.
Stochastic conduction block	Multiplicative, temporally persistent Bernoulli dropout	To be completed	Must be modeled as all-or-none, burst-like dropout—not additive Gaussian noise.
Consolidated lesion matrix	Amplitude, delay and both $\times$ uniform and non- uniform; seeded; two-stream MT	To be completed	Targeted cells are complete, but the remaining pre-mtMix conditions need one consolidated pass with Figs. 9–14 and motion-letter d' .